

EXP2-2 – Twists and Turns

The **aim** of EXP2-2 is to consider how to plan a set of measurements to determine the shear modulus of materials from the twisting of a wire.

The **objectives** for Part 2 of this exercise are:

- Collect the experimental data based on your experimental plan
- Calculate the shear modulus and damping factor
- Consider the uncertainty of measurement in the results for the shear modulus
- Critically discuss the results
- Present a short analytical report

What you should do:

In your groups, use the data you collect to calculate the shear modulus of the metal wire and the shear modulus and damping behaviour (Q and $\tan\delta$) of the polymer wire.

- Plot T^2 against L for both the metal wire and the polymer wire
- Plot $\ln A_0 / A_n$ against n for the polymer wire
- From these graphs and the measurements of the other parameters in the experiment calculate a value for the shear modulus of the 2 wires and the quality factor and $\tan\delta$ of the polymer wire.
- Estimate the uncertainties in each measurement.
- Make an uncertainty budget table for the measurements.
- State the values for the shear moduli and damping behaviour together with the uncertainty in the values to the 95% confidence level.

Write a report on your experimental measurements.

The report should contain the following sections:

1. Introduction:

Briefly explain what the experiment is trying to achieve and how you can obtain the shear modulus and damping factor from a torsional pendulum. Why is it a better method than calculating shear modulus from the results of a tensile test?

2. Methods:

You do NOT need to describe the procedure of the experimental methods, refer to the “Experimental plan” you submitted on QMPlus.

3. Results:

Present the results and calculations.

Use tables or figures with correct axes, units and captions. (Including 3 plots, the equations you’ve used to calculate the uncertainties, and the final results expressed in scientific format)

Complete your uncertainty budget table.

For detailed calculation process, you can refer to the excel file you will submit together with your report.

4. Discussion:

Discuss the results that you obtain:

- Are the methods good enough to make precise measurements? (Please review your understanding of statistics to help you determine measurement precision)
- Where do the largest uncertainties come from? (Refers to your uncertainty budget table) What could you do to reduce them?
- Given that the metal wire is a type of alloy steel with some Carbon and the polymer wire is a type of nylon (polyamide 66) – comment on whether the results from the experiment are accurate.

5. Conclusion:

What can you conclude from the experiment results?

(Give a summary of your results and discussion)

6. Reflection:

Reflect on what you think you have learned about laboratory measurement from this exercise and any other personal reflections.

The report will be **no more than 1600 words and not more than 4 sides of A4 paper excluding names and references**. Page margins may not be less than 2cm, line spacing no less than 1.0 and letters no smaller than font size 11.

The **title** of the report will be: **EXP2-2 Twists and Turns Report XX** (where XX is your group number, e.g. M16 or P8)

The names of **the group members** should be given as authors of the report.

You will **submit** your group's report as a **pdf document on QMPlus** using the file name: **'QXU5017 EXP2-2 Part 2 Report XX.pdf'** (where XX is your group number, e.g. M9 or P19).

You will also need to submit your raw data spreadsheet (must include your readings) file including all the calculation process using the file name: **'QXU5017 EXP2-2 Data XX'**.

The report must be completed and **submitted by 24:00 on Friday 28th April 2025**.

The report must be the **work of the whole group** and the content agreed by all group members.

Only one group member needs to hit the submit button to submit the report.

Marks breakdown:

EXP2-2 is 30% of the marks for QXU5017

Part 1 Group Plan 1 will be worth 40% of EXP2-2

Part 2 Group Report 2 will be worth 60% of EXP2-2